A SHEAF OF ABELIAN GROUPS ON THE BIG ETALE SITE VANISHING BELOW DIMENSION N, NONZERO FROM DIMENSION N, INVARIANT UNDER THICKENINGS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof is 'See the discussion above'; the context carries the complete n=1 verification and the G_n variant is derived by the author's explicit one-shows analogy step - recorded as mechanical analogy, content verbatim.

Editorial note. The label lemma-sheaf-zero-on-low-dimension names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 4517-4604: F(X) = product over pairs of distinct points with x' leading to x of Z/2Z, the contravariant F(f), sheafification G; the three verifications - G(X) != 0 when a nontrivial specialization exists (generalizations lift along flat morphisms, Morphisms lemma-generalizations-lift-flat), G(X) = 0 when no specializations (fields and more generally rings with all primes maximal), G(X) = G(X') for thickenings (topological invariance of the etale site, Etale Cohomology theorem-topological-invariance); the variant F_n on |X|^{n+1} with sheafification G_n
- More on Morphisms, definition-thickening

Source reference. Original theorem 04DZ: sources/stacks-chow/tags.txt, source lines 5745–5745. Prerequisite pointer invoked at 4693; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 04EX: sources/stacks-chow/tags.txt, source lines 5778–5778. Prerequisite pointer invoked at 4688; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Prerequisite pointer invoked at 4674; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 021A: sources/stacks-chow/tags.txt, source lines 2663–2663. Prerequisite pointer invoked at 4622; it names a standard fact of

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a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

In the following we fix a big étale site $Sch_{étale}$ as constructed in Topologies, Definition 021A. Moreover, a scheme will be an object of this site. Recall that if x, x' are points of a scheme X we say x is a *specialization* of x' or we write $x' \rightsquigarrow x$ if $x \in \overline{\{x'\}}$. This is true in particular if $x = x'$.

Consider the functor $F : Sch_{étale} \to Ab$ defined by the following rules:

$$F(X) = \prod\nolimits_{x \in X} \prod\nolimits_{x' \in X, x' \rightsquigarrow x, x' \neq x} \mathbf{Z}/2\mathbf{Z}$$

Given a scheme X we denote $|X|$ the underlying set of points. An element $a \in F(X)$ will be viewed as a map of sets $|X| \times |X| \to \mathbf{Z}/2\mathbf{Z}$, $(x, x') \mapsto a(x, x')$ which is zero if $x = x'$ or if x is not a specialization of x'. Given a morphism of schemes $f : X \to Y$ we define

$$F(f) : F(Y) \longrightarrow F(X)$$

by the rule that for $b \in F(Y)$ we set

$$F(f)(b)(x, x') = \begin{cases} 0 & \text{if } x \text{ is not a specialization of } x' \\ b(f(x), f(x')) & \text{else.} \end{cases}$$

Note that this really does define an element of $F(X)$. We claim that if $f : X \to Y$ and $g : Y \to Z$ are composable morphisms then $F(f) \circ F(g) = F(g \circ f)$. Namely, let $c \in F(Z)$ and let $x' \rightsquigarrow x$ be a specialization of points in X, then

$$F(g \circ f)(x, x') = c(g(f(x)), g(f(x'))) = F(g)(F(f)(c))(x, x')$$

because $f(x') \rightsquigarrow f(x)$. (This also works if $f(x) = f(x')$.)

Let G be the sheafification of F in the étale topology.

I claim that if X is a scheme and $x' \rightsquigarrow x$ is a specialization and $x' \neq x$, then $G(X) \neq 0$. Namely, let $a \in F(X)$ be an element such that when we think of a as a function $|X| \times |X| \to \mathbf{Z}/2\mathbf{Z}$ it is nonzero at (x, x'). Let $\{f_i : U_i \to X\}$ be an étale covering of X. Then we can pick an i and a point $u_i \in U_i$ with $f_i(u_i) = x$. Since generalizations lift along flat morphisms (see Morphisms, Lemma 03HV) we can find a specialization $u_i' \rightsquigarrow u_i$ with $f_i(u_i') = x'$. By our construction above we see that $F(f_i)(a) \neq 0$. Hence a determines a nonzero element of $G(X)$.

Note that if $X = \mathrm{Spec}(k)$ where k is a field (or more generally a ring all of whose prime ideals are maximal), then $F(X) = 0$ and for every étale morphism $U \to X$ we have $F(U) = 0$ because there are no specializations between distinct points in fibres of an étale morphism. Hence $G(X) = 0$.

Suppose that $X \subset X'$ is a thickening, see More on Morphisms, Definition 04EX. Then the category of schemes étale over X' is equivalent to the category of schemes étale over X by the base change functor $U' \mapsto U = U' \times_{X'} X$, see Étale Cohomology, Theorem 04DZ. Since it is always the case that $F(U) = F(U')$ in this situation we see that also $G(X) = G(X')$.

As a variant we can consider the presheaf F_n which associates to a scheme X the collection of maps $a : |X|^{n+1} \to \mathbf{Z}/2\mathbf{Z}$ where $a(x_0, \ldots, x_n)$ is nonzero only if $x_n \rightsquigarrow \ldots \rightsquigarrow x_0$ is a sequence of specializations and $x_n \neq x_{n-1} \neq \ldots \neq x_0$. Let G_n be the sheaf associated to F_n. In exactly the same way as above one shows that G_n is nonzero if $\dim(X) \geq n$ and is zero if $\dim(X) < n$.

Lemma 0.1. *There exists a sheaf of abelian groups G on $Sch_{étale}$ with the following properties*

(1) $G(X) = 0$ *whenever* $\dim(X) < n$,

(2) $G(X)$ *is not zero if* $\dim(X) \geq n$, *and*
(3) *if* $X \subset X'$ *is a thickening, then* $G(X) = G(X')$.

Proof. See the discussion above. □